CS & DA

Python-For Data Science

Python Datatypes

DPP: 1

Q1 Output Printed by below Code is _____

```
Set1={2,4,6,8,10}
Set2={1,2,3,4,5}
Set3={1,3,5,7,9,1,5}
s1=Set1 & Set2
s2=Set2 | Set3
x=len(Set3)
for i in s1:
   for j in s2:
      x=x+1
print(x)
```

(A) 13 (B) 19
(C) 23 (D) 29

Q2 The number of times print statement is executed is _____

```
s1={5,7,9,7,5}
s2={3,5,7,5,3}
s3={1,2,3,4,5}
for i in range(len(s1.difference(s3))):
    for j in range(len(s3.union(s2))):
        print("Hi")
```

Q3 The output of below code will be _____.

```
d1={'a':1,'b':2,'c':3}
d2={'c':4,'d':5,'e':6}
d1.update(d2)
print(d1)
```

(A) {'a': 1, 'b': 2, 'c': 3, 'd': 5, 'e': 6}
(B) {'a': 1, 'b': 2, 'c' : 3,'c': 4, 'd': 5, 'e': 6}
(C) {'a': 1, 'b': 2, 'c': 4, 'd': 5, 'e': 6}
(D) {'a': 1, 'b': 2, 'd': 5, 'e': 6}

Q4 The final count value will be _____

```
myList=[11,22,33,44,55,66]
newList=[item for item in myList if item%2==0]
count=1
for i in range(len(newList)):
   count+=newList[i]
```

Q5 What is printed by below code?

```
a = {'apple','banana','mango'}
b = {'mango','grapes','chiku'}
c = {'chiku','banana','guava'}
x = len(a–b)
y = len(b^c)
z = len(c&a)
print(x+y+z)
```

(A) 6 (B) 7
(C) 8 (D) 9

Q6 The total number of times print statement executed is _____

```
a = [10,20,[30,40,50],60,70]
b = (5,10,15,(20,25,30,(35,40,45,50),55,60),70)
c = a[2][1]
d = b[3][3][3]
i = 1
for j in (c,1,c//4):
   for k in (d,1,c//5):
      print(i)
```

Q7 The output will be _____

```
x = "GATE"
y = "2025"
z = 'EXAM'
string = x + y + z
print(len(string*2))
```

(A) 4 (B) 8
(C) 16 (D) 24

Q8 The final value of ans will be _____

```
i = {10,20,30,40,50}
j = [11,22,33,44,55]
k = [22,33,44,11,55]
l = {30,20,40,10,50}
x = i  == l
y = j != k
print(x,y)
```

(A) False, False (B) True, False (C) False, True (D) True, True

Answer Key

Q1	B
Q2	12
Q3	C
Q4	133
Q5	B
Q6	10
Q7	D
Q8	D

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

- s1 = {2, 4}
- s2 = {1, 2, 3, 4, 5, 7, 9}

There are 2 elements in s1 and 7 elements in s2.
The nested loops iterate as follows:

- For each element i in s1 (2 iterations):
  - For each element j in s2 (7 iterations per i):
    - x = x + 1

Total iterations: $2 \times 7 = 14$2×7=14

Final Value of x

- Initially, x = 5
- During the nested loops, x is incremented 14 times

So the final value of x is:
$x = 5 + 14 = 19$x=5+14=19

Output

The output printed by the code is 19

Q2 Text Solution:

1. **Sets Initialization**
   - s1 = {5, 7, 9, 7, 5}
     Note: Sets automatically remove duplicates, so s1 becomes {5, 7, 9}.
   - s2 = {3, 5, 7, 5, 3}
     Similarly, s2 becomes {3, 5, 7}.
   - s3 = {1, 2, 3, 4, 5}
2. **Set Operations**
   - **Difference (s1.difference(s3))**
     - s1.difference(s3) finds elements in s1 that are not in s3.
     - s1 = {5, 7, 9}
     - s3 = {1, 2, 3, 4, 5}
     - The difference will be {7, 9} because 5 is in s3, so only 7 and 9 remain.
   - **Union (s3.union(s2))**
     - s3.union(s2) finds all unique elements in either s3 or s2.
     - s3 = {1, 2, 3, 4, 5}
     - s2 = {3, 5, 7}
     - The union will be {1, 2, 3, 4, 5, 7}.

Calculations

1. **Length of s1.difference(s3)**
   - The difference is {7, 9}.
   - The length of this set is 2.
2. **Length of s3.union(s2)**
   - The union is {1, 2, 3, 4, 5, 7}.
   - The length of this set is 6.

The total number of times print("Hi") is executed is calculated by multiplying the iterations of the outer loop by the iterations of the inner loop:
$\text{Total executions} = 2 \times 6 = 12$
Total executions=2×6=12
Thus, the print("Hi") statement is executed 12 times

Q3 Text Solution:

1. **Initial Dictionaries**:
   - d1: {'a': 1, 'b': 2, 'c': 3}
   - d2: {'c': 4, 'd': 5, 'e': 6}
2. **Updating d1 with d2**:
   - The key 'c' is present in both d1 and d2. The value in d2 (4) will overwrite the value in d1 (3).
   - Keys 'd' and 'e' from d2 will be added to d1 because they do not exist in d1.

Resulting Dictionary

After the update() operation, d1 will be:

- 'a': 1 (unchanged)
- 'b': 2 (unchanged)
- 'c': 4 (updated value from d2)
- 'd': 5 (added from d2)
- 'e': 6 (added from d2
-
- CORRECT -C

Q4 Text Solution:

1. Let's compute the value of count during each iteration:
   - **Initial count:** 1
   - **Iteration 1 (i = 0):**
     - newList[0] is 22
     - Update count: count += 22 count = 1 + 22 = 23
   - **Iteration 2 (i = 1):**
     - newList[1] is 44
     - Update count: count += 44 count = 23 + 44 = 67
   - **Iteration 3 (i = 2):**
     - newList[2] is 66
     - Update count: count += 66 count = 67 + 66 = 133

Final count Value

After all iterations, the final value of count is 133.
Thus, the final count value will be **133**.

Q5 Text Solution:

- c & a is the intersection between sets c and a, which includes elements common to both sets.
- c = {'chiku', 'banana', 'guava'}
- a = {'apple', 'banana', 'mango'}
- The common element is {'banana'}.

z = len({'banana'}) # length is 1
z = len({'banana'}) # length is 1
x + y + z = 2 + 4 + 1 = 7

Q6 Text Solution:

1. **Determine c and d Values**
   - a = [10, 20, [30, 40, 50], 60, 70]
     - a[2] is [30, 40, 50]
     - a[2][1] is 40
     - Therefore, c = 40
   - b = (5, 10, 15, (20, 25, 30, (35, 40, 45, 50), 55, 60), 70)
     - b[3] is (20, 25, 30, (35, 40, 45, 50), 55, 60)
     - b[3][3] is (35, 40, 45, 50)
     - b[3][3][3] is 50
     - Therefore, d = 50
2. **Calculate the Tuples for Loop Iteration**
   - **First Loop: j Values**
     - c = 40
     - 1 = 1
     - c // 4 = 40 // 4 = 10
     - So, j takes values in (40, 1, 10)
   - **Second Loop: k Values**
     - d = 50
     - 1 = 1
     - c // 5 = 40 // 5 = 8
     - So, k takes values in (50, 1, 8)
3. **Calculate Iterations**
   - **Outer Loop**: Iterates over j values: 40, 1, 10 — total of 3 iterations.
   - **Inner Loop**: Iterates over k values: 50, 1, 8 — total of 3 iterations.

The total number of iterations of the print(i) statement is:

$$\text{Number of iterations} = (\text{number of } j \text{ values}) \times (\text{number of } k \text{ values})$$

Number of iterations=

(number of j values)×(number of k values)

$$\text{Number of iterations} = 3 \times 3 = 9$$

Number of iterations=3×3=9+1

Thus, the print(i) statement is executed 10 times.

Q7 Text Solution:

1. string = "GATE2025EXAM"
2. string * 2 = "GATE2025EXAMGATE2025EXAM"
3. len("GATE2025EXAMGATE2025EXAM") = 24

The correct output should be 24.

Q8 Text Solution:

- x = True
- y = True

Output

The final output of the print(x, y) statement is:
(D) True, True

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